SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

*I **Y.GEETHA REDDY(192572053)(**Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:*Geetha*

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _______→ Transport → Network → _______→ Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

Application → Presentation → Session → Transport → Network → Data Link → Physical

Explanation

- Session Layer: Establishes, manages, and terminates communication sessions between devices. It also provides synchronization and recovery during data transfer.
- Data Link Layer: Converts packets into frames, adds MAC addresses, detects errors, and ensures reliable communication between directly connected devices.

Both layers are essential for reliable communication because the Session Layer maintains communication continuity, while the Data Link Layer ensures error-free frame delivery.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

Application
|
Presentation
|
Session
|
Transport
|
Segment
|
Network
|
Packet
|
Data Link
|
Frame
|
Physical
|
Bits

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✔ → Data Link ✔ → Network ✖ → Transport ✖ → Application ✖

Analyze the problem and explain how the concept map helps in troubleshooting.

4. A concept map shows:

ErrorControl data

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

Explanation

- The Transport Layer converts data into Segments.
- The Network Layer converts segments into Packets.
- The Data Link Layer converts packets into Frames.
- The Physical Layer converts frames into Bits for transmission.

This process is called encapsulation, which ensures that each layer adds the required information for successful communication

3.Given the concept

DataLink→MACAddress→LocalDelivery Network → IP

Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

```
Data Link Layer
      |
  MAC Address
      |
 Local Delivery

Network Layer
      |
  IP Address
      |
 Global Delivery
```

Explanation

- The MAC Address identifies devices within the same local network and is used for local delivery.
- The IP Address identifies devices across different networks and enables global communication through routers.
- Together, MAC and IP addresses ensure that data reaches the correct destination both locally and globally.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in transmission.

```
Error Control
    |
 ┌─────────────────────┐
 |          |
Data Link Layer  Transport Layer
    |          |
Error Detection  Error Correction
```

Explanation

- The **Data Link Layer** detects transmission errors using CRC or checksum.
- The **Transport Layer** ensures reliable communication by retransmitting lost or corrupted data using acknowledgements and sequence numbers.
- Together, these layers improve reliability by detecting, correcting, and recovering from transmission errors.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

```
User
  |
Application (HTTP/HTTPS)
  |
Presentation (Encryption, SSL/TLS)
  |
Session (Session Management)
  |
Transport (TCP)
  |
Network (IP)
  |
Data Link (Ethernet/Wi-Fi)
  |
Physical (Cables/Wireless)
```

Explanation

- The Application Layer handles web requests.
- The Presentation Layer encrypts and decrypts data.
- The Session Layer manages communication sessions.
- The Transport Layer provides reliable data transfer.
- The Network Layer routes packets.
- The Data Link Layer delivers frames within the local network.
- The Physical Layer transmits bits through cables or wireless media.

Effect of Layer Failure

- If any one layer fails, communication is interrupted.
- For example, if the Network Layer fails, packets cannot reach the destination, causing the webpage to fail to load.

6. A network issue is represented in the concept map:

Physical ✔ → Data Link ✔ → Network ✖ → Transport ✖ → Application ✖

Analyze the problem and explain how the concept map helps in troubleshooting.

```
Physical ✔
     |
Data Link ✔
     |
Network ✖
     |
Transport ✖
     |
Application ✖
```

Explanation

- The Physical Layer is working correctly, so cables or wireless connections are functioning.
- The Data Link Layer is also working, indicating that frames are being transmitted successfully within the local network.
- The Network Layer has failed, meaning IP routing or addressing is not functioning.
- Since the Network Layer has failed, the Transport Layer cannot establish end-to-end communication.
- As a result, the Application Layer also fails because no data reaches the application.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation